

Antiretroviral Therapy

Introduction

- ❖ Substantial advances has made after zidovudine discovery (1987)
- ❖ Currently 6 classes of antiretroviral agents are available
 - nucleoside/nucleotide reverse transcriptase inhibitors (NRTIs),
 - non nucleoside reverse transcriptase inhibitors (NNRTIs),
 - protease inhibitors (PIs),
 - fusion inhibitors,
 - CCR5 co-receptor antagonists, and
 - integrase strand transfer inhibitors (InSTIs)
 - These agents inhibit HIV replication at different parts of the cycle

Introduction

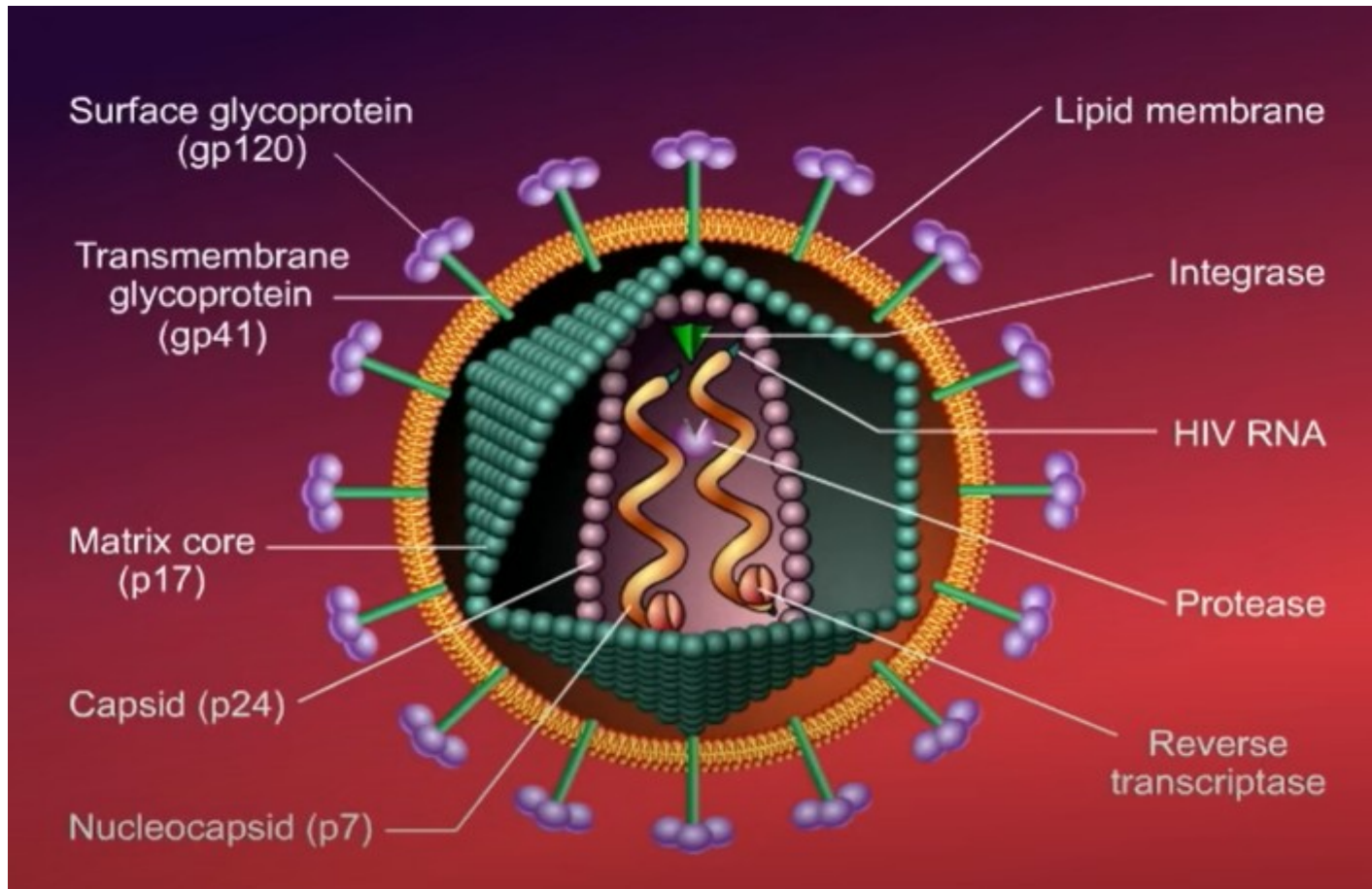
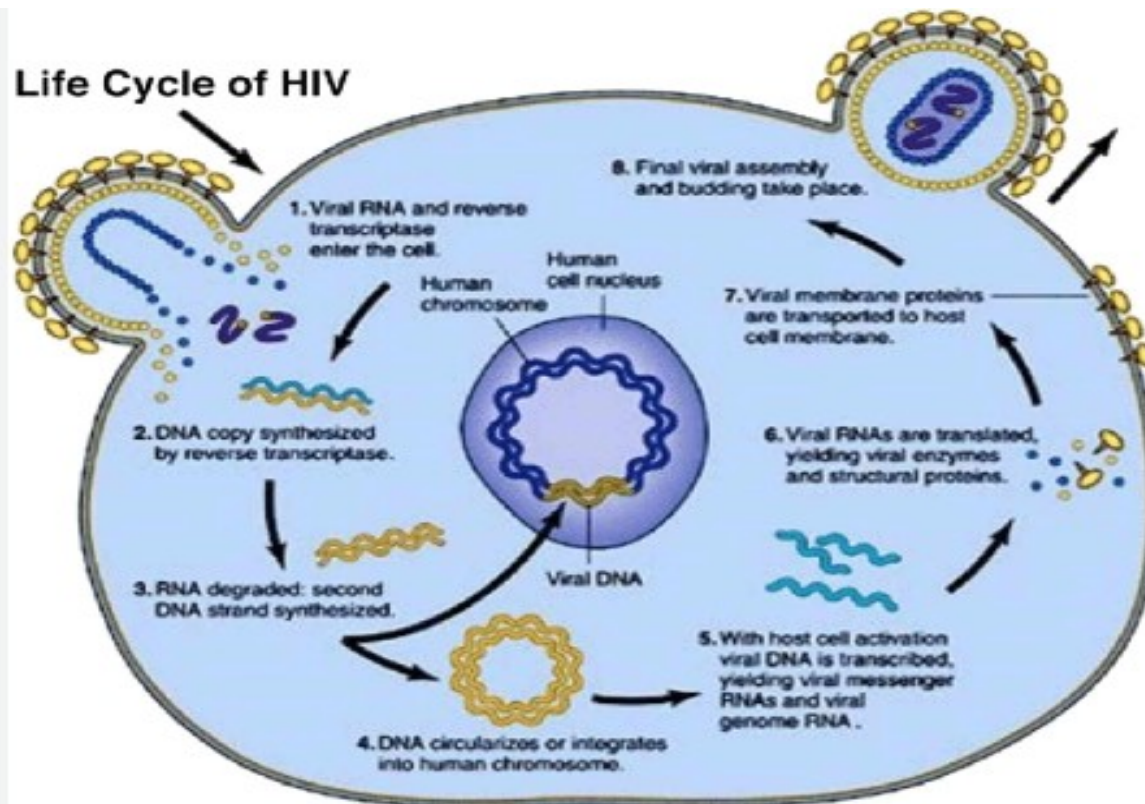


Figure: Structure of HIV virus



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Figure: Life cycle of HIV

Introduction

The emergence of **testing virus through** viral load and resistance testing enables the use of combination therapy with maximally potent agents

- this **reduce viral replication** to the lowest possible level,
- decreasing the likelihood of **emergence of resistance**.

❖ administration of combination antiretroviral therapy has become the **standard of care**.

❖ Viral susceptibility to specific agents varies among patients and may change with time.

❖ **Importance factors for selections of Drugs**

- potency and susceptibility,
- Convenience,
- optimization of adherence, and
- drug-drug interactions

Introduction

- ❖ Older drugs use now diminished. These drugs are
- ❖ delavirdine, didanosine (ddI), indinavir, nelfinavir, and stavudine (d4T). Zalcitabine (ddC; dideoxycytidine)
 - no longer marketed.

Table: drugs recommended during pregnancy

Recommended Agents	Alternate Agents
Nucleoside/nucleotide reverse transcriptase inhibitors (NRTIs)	
Abacavir, emtricitabine, lamivudine, tenofovir disoproxil fumarate, zidovudine	
Non-nucleoside reverse transcriptase inhibitors (NNRTIs)	
Efavirenz	Rilpivirine
Protease inhibitors (PIs)	
Atazanavir/ritonavir, darunavir/ritonavir	Lopinavir/ritonavir
Integrase inhibitors	
Raltegravir	

Nucleotide Reverse Transcriptase Inhibitors (NRTIs)

GENERIC NAME	ABBREVIATION; Chemical Name
Nucleoside/Nucleotide Reverse Transcriptase Inhibitors	
Zidovudine ^a	ZDV; azidothymidine (AZT)
Didanosine	ddI; dideoxyinosine
Stavudine	d4T; didehydrodeoxythymidine
Zalcitabine ^b	DDC; dideoxycytidine
Lamivudine ^a	3TC; dideoxythiacytidine
Abacavir ^a	ABC; amino-cyclopropylamino-purinylcyclopentenylmethanol
Tenofovir disoproxil ^{a,c} Tenofovir alafenamide ^{a,c}	TDF; TAF; phosphono-methoxypropyladenine (PMPA)
Emtricitabine ^a	FTC; fluorooxathiolanyl cytosine

Nucleotide Reverse Transcriptase Inhibitors (NRTIs)

MOA: act by competitive inhibition of HIV-1 reverse transcriptase;

—causes **premature chain** termination due to inhibition of binding with the incoming nucleotide

❖ Each agent requires intracytoplasmic activation via phosphorylation by cellular enzymes to the **triphosphate form**.

❖ **Mutations for resistance** include M184V, K65R/K, and K70E..

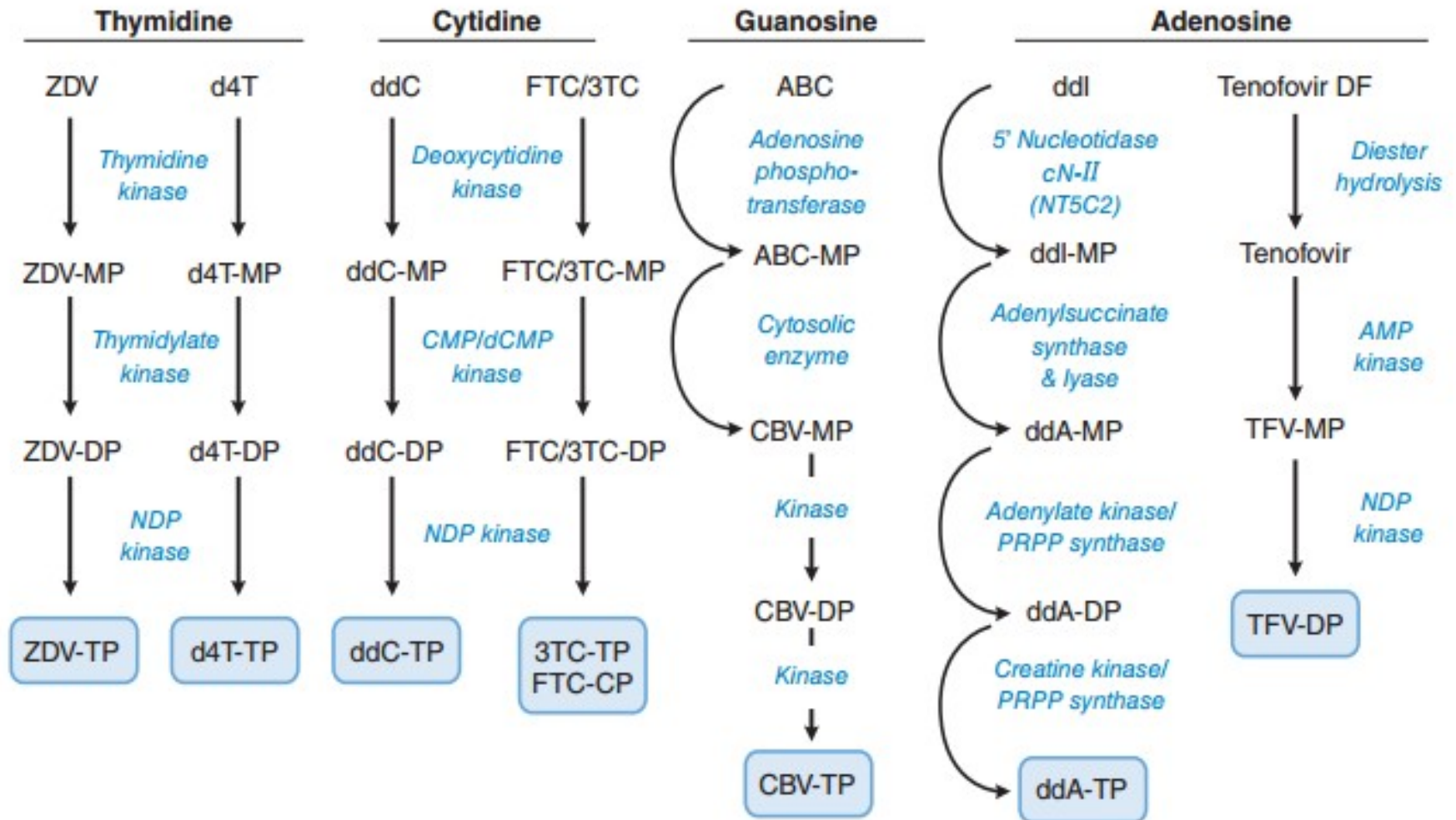


Fig: Intracellular activation of nucleoside analogue reverse transcriptase inhibitors.

Toxicities of NRTIs

- ❖ All NRTIs may be associated with mitochondrial toxicity

Manifestations of toxicities

- peripheral neuropathy,
 - pancreatitis,
 - lipoatrophy, or hepatic steatosis.
- ❖ Less commonly, lactic acidosis (can be fatal). NRTI treatment should be suspended
 - in the setting of rapidly rising aminotransferase levels,
 - progressive hepatomegaly, or
 - metabolic acidosis of unknown cause.

Toxicities of NRTIs

- ❖ **Lipoatrophy and insulin resistance occur most frequently** with thymidine analogs
 - Stavudine
 - zidovudine,
- ❖ and least frequently with use of
 - tenofovir,
 - lamivudine,
 - emtricitabine, and
 - abacavir.

Abacavir

- It is a guanosine analog agent

PK

- ❖ well absorbed following oral administration (83%) and is
- ❖ **Its effect is unaffected by food.**
- ❖ Half-life.....1.5 hours.
- ❖ **Metabolism:** hepatic glucuronidation and carboxylation.

In case of liver disease

- **Dosage reduction** is recommended in **mild hepatic impairment;**
- **avoidance of use is recommended in patients with moderate or severe liver disease.**
- ❖ Concomitant alcohol use might increase serum levels of abacavir
 - **Metabolized by ADH**
- ❖ recommended for use in pregnancy

ADR & Toxicities

- ❖ **Hypersensitivity reactions (8%)occasionally fatal,**
 - may be more severe with once-daily dosing.**
- ❖ Symptoms occurs in the **first 6 weeks** of therapy but less frequent
 - fever, fatigue, nausea, vomiting, diarrhea, and abdominal pain.
- ❖ Other:
- ❖ Dyspnea, pharyngitis, cough, and elevations in serum aminotransferase or creatine
- ❖ Screening for **HLA-B*5701** before initiation of abacavir therapy is important
 - Helps to identify identify patients with an increased risk for abacavir-associated hypersensitivity reaction**

ADR & Toxicities

mitochondrial toxicity and disorders of lipid metabolism

– less common with **abacavir** than with other nucleoside analogs.

Didanosine (ddI)

❖ It is a synthetic analog of deoxyadenosine.

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✓ bioavailability40%

✓ Half-life..... 1.5 hours

✓ Elimination

–eliminated by both cellular metabolism and renal excretion.

❖ Formulation should be **Buffered** or **enteric-coated**

–to prevent inactivation by gastric acid.

❖ The major clinical toxicities

✓ distal sensory neuropathy and

✓ dose-dependent pancreatitis

–coadministration with drugs or conditions that increase the risk of either (eg, alcohol abuse, hypertriglyceridemia, pregnancy, zalcitabine, stavudine, isoniazid, vincristine, ribavirin, and hydroxyurea) should be avoided.

Other reported adverse effects

–Diarrhea,

–hepatitis,

–esophageal ulceration,

–cardiomyopathy,

–central nervous system toxicity (headache, irritability, insomnia), and

–hypertriglyceridemia.

ADR & Toxicities of Didanosine...

- ❖ Concurrent stavudine administration increases the risk of **lactic acidosis**.
- ❖ Reports of **retinal changes** and **optic neuritis**
 - particularly in adults receiving high doses and in children
 - mandate periodic retinal examinations
- ❖ **tenofovir**increase ddi
 - Dose should be reduce.
- ❖ Concurrent **ribavirin** is C/I.
- ❖ The chewable tablets **contain phenylalanine**, be
 - harmful to patients **with phenylketonuria**.

Emtricitabine (FTC)

It is a fluorinated analog of lamivudine with a long intracellular

PK

- ✓ half-life (>24 hours),
- ✓ Oral bioavailability of the capsules....**93% and is unaffected by food,**
- ✓ but penetration into the cerebrospinal fluid is low.
- ✓ **Elimination** is by both **glomerular filtration** and active **tubular secretion.**
- ❖ FTC is one of the NRTI agents recommended for use in pregnancy

Emtricitabine (FTC)...

- ❖ FTC + tenofovir disoproxil fumarate
 - FDA-approved formulation for pre-exposure prophylaxis to reduce HIV acquisition in high-risk persons
- ❖ Both **emtricitabine** and **lamivudine** have
 - M184V/I** mutation and therefore should not be used together.

Adverse effects

- ❖ **The most common adverse effects**
 - headache, diarrhea, nausea, and rash.
- ✓ Hyperpigmentation of the palms or soles may occur (~3%)
- ✓ Due to its activity against HBV,
 - exacerbation of **HBV** may occur if therapy is interrupted or discontinued in patients co-infected with HIV.

- ❖ Lamivudine (3TC) is a **cytosine** analog
- ❖ It has in vitro activity against both **HIV-1** and **HBV**.
- ❖ Lamivudine is one of the recommended **NRTI** agents for use in **pregnant women**

PK

- Oral bioavailability > 80% and is **not food-dependent**.
- Serum half-life..... 2.5 hours,
- intracellular half-life of the triphosphorylated compound ... 11–14 hours.
- predominantly eliminated in the urine by **active organic cation secretion**.

Adverse effects.....uncommon but includes

- but include headache, dizziness, insomnia, fatigue, dry mouth, and gastrointestinal discomfort.

Lamivudine...

- ❖ Due to its activity **against HBV**,
- ❖ if therapy is interrupted or discontinued in patients co-infected with HIV and HBV....HBV would be exacerbated
- ❖ **3TC and FTC** should not be used together
 - M184V mutation has a high level of resistance for both agents

Drug interaction

- ❖ TMP-SMX....(+) levels of 3TC
- ❖ **Lamivudine** and **zalcitabine** may inhibit the intracellular **phosphorylation of one another**;
 - their concurrent use should be avoided if possible.

Stavudine (d4T) is thymidine analog

PK

- oral bioavailability (86%) that is not food-dependent.
- The serum half-life 1.1 hours

Adverse effects

❖ **major toxicity.....dose-related peripheral sensory neuropathy**

- increased with concomitant neurotoxic drugs

❖ **Other potential adverse effects are**

- pancreatitis, arthralgias, and elevation in serum aminotransferases.
 - Caution is advised in patients with liver **dysfunction**.

❖ **A rare adverse effect**

- rapidly progressive ascending neuromuscular weakness.

❖ **Lactic acidosis with hepatic steatosis, as well as lipodystrophy,**

- more frequent in d4T than in those receiving other NRTI agents.

Stavudine

Drug interaction

- ❖ d4T with DDI should not be administered
 - due to increased risk of **both lactic acidosis and pancreatitis.**
 - This combination has been implicated in several deaths in **HIV-infected pregnant women.**
- ❖ Zidovudine with d4T should not be used together
 - Zidovudin may reduce the **intracellular phosphorylation** of d4T

Tenofovir Disoproxil Fumarate

- ❖ Tenofovir is an acyclic nucleoside phosphonate (ie, nucleotide) analog of adenosine
- ❖ Tenofovir disoproxil fumarate is a water-soluble prodrug of active tenofovir
- ❖ recommended for use in pregnancy

Spectrum of activity

- ❖ activity against HIV and HBV.
- ❖ only two rather than three **intracellular phosphorylations** are required for active inhibition of DNA synthesis.

PK

- High fat meal increases the oral bioavailability (from 25 % to 39)
- Prolonged half life (12–17 hours)
- Elimination occurs by both **glomerular filtration** and **active tubular secretion**,
 - dosage adjustment in patients with renal insufficiency is recommended.

Tenofovir Disoproxil Fumarate

- ❖ Combinations of tenofovir disoproxil fumarate and FTC
 - The only **FDA-approved formulation** for pre-exposure prophylaxis to reduce HIV acquisition in high-risk persons.

Averse Effects

- ❖**The most common....**Gastrointestinal complaints (eg, nausea, diarrhea, vomiting, flatulence) but rarely require discontinuation.
- ❖**lactose intolerance**increased with individuals of this condition
 - Since **tenofovir** is formulated **with lactose**, these may occur more frequently in patients with lactose intolerance.
- ❖**Cumulative loss of renal function**
 - increased with concurrent use of **boosted PI regimens** or in patients with pre-existing renal disease

Averse Effects

- ❖ Acute renal failure and **Fanconi syndrome** have also been reported.
 - should be used with caution in patients at risk for renal dysfunction.
 - Serum creatinine levels should be monitored during therapy and
 - should be D/C discontinued for new proteinuria, glycosuria, or $\text{CrCl} < 30 \text{ mL/min}$.
- ❖ Tenofovir-associated proximal **renal tubulopathy** causes
 - Excessive renal phosphate and calcium losses and 1-hydroxylation defects of vitamin D
- ❖ Tenofovir is not recommended in individuals with or at risk for **bone demineralization**.
 - loss of bone mineral density and osteomalacia have been reported,

Drug interaction

- ❖ **Tenofovir may compete with** other drugs that are actively secreted by the kidneys
 - Example.....cidofovir, acyclovir, and ganciclovir
- ❖ **Concurrent use of probenecid is C/I.**
- ❖ **Tenofovir + telaprevir**
 - Tenofovir levels may increase, and levels of **telaprevir decrease**

- ❖ Tenofovir alafenamide is a phosphonoamidate prodrug of tenofovir
- ❖ available only in **co-formulation** with other antiretroviral agents (with **emtricitabine**, with elvitegravir plus **cobicistat** plus emtricitabine, and with **rilpivirine plus emtricitabine**).
- ❖ **Compared to tenofovir disoproxil,**
 - ✓ Plasma levels of active tenofovir in plasma lower with tenofovir alafenamide than with tenofovir disoproxil,
 - metabolism occurs in lymphocytes and macrophages (as well as hepatocytes and some other cells) rather than blood
 - ✓ Tenofovir alafenamide has **less renal and bone toxicity**

Tenofovir alafenamide

❖ Adverse effects

✓ **uncommon** but may include gastrointestinal symptoms or headache.

❖ Tenofovir alafenamide has also been approved for treatment of HBV infection

Drug interaction

❖ Tenofovir alafenamide is a substrate of **P-glycoprotein**,

–levels of tenofovir can be affected by inhibitors or inducers of P-glycoprotein.

❖ Ritonavir and cobicistat(+) plasma concentrations of tenofovir alafenamide, while **darunavir** can decrease (-)

- ❖ Zidovudine (**azidothymidine**; AZT) is a deoxythymidine analog that
- ❖ **Zidovudine** was the first antiretroviral agent to be approved and has been well studied
- ❖ Produce significant reductions in the rate of vertical transmission during pregnancy, labor, and postpartum
 - one of the NRTI agents recommended during pregnancy
- ❖ Used as an option for **postexposure prophylaxis** in individuals exposed to HIV

Zidovudine...

PK

- ❖ It is well absorbed (63%) and distributed to most body tissues and fluids, including the CSF,
- ❖ **half-life**1 hour but the **intracellular half-life** of the phosphorylated compound is 3–4 hours,
 - twice-daily dosing is sufficient
- ❖ **eliminated primarily** by renal excretion following **glucuronidation** in the liver.

Zidovudine...

Adverse effects

❖ The most common adverse effects

- macrocytic anemia (1–4%) and
- neutropenia (2–8%).

❖ **Adverse effects that may occur**

✓ Gastrointestinal intolerance, headaches, and insomnia

- but tend to resolve during therapy and

❖ A symptomatic **myopathy** may occur with prolonged use.

❖ **Lipoatrophymore common** in patients receiving zidovudine or other thymidine analogs.

❖ **High doses** can cause anxiety, confusion, and tremulousness.

Zidovudine...

Drug interaction

- ❖ Induction or inhibition of glucuronidation may alter serum levels of zidovudine
 - ✓ when coadministered with atovaquone, lopinavir/ritonavir, probenecid, or valproic acid.
- ❖ **AZT + D4t.....C/I**
 - due to competitive inhibition of intracellular phosphorylation.

Nonnucleoside Reverse Transcriptase Inhibitors (NNRTIs)

Drugs in this class: Nevirapine, Efavirenz, Delavirdine , Etravirine & Rilpivirine

MOA : The NNRTIs bind directly to HIV-1 reverse transcriptase

- resulting in allosteric inhibition of RNA- and DNA-dependent DNA polymerase activity.
- The binding site of NNRTIs is near to but distinct from that of NRTIs.
- ❖ Unlike the NRTI agents, NNRTIs **neither compete** with nucleoside triphosphates **nor require phosphorylation to be active.**
- ❖ The second-generation **NNRTIs** (doravirine, etravirine, rilpivirine)
 - have higher potency, longer half-lives and reduced side-effect profiles compared with older NNRTIs (delavirdine, efavirenz, nevirapine).

ADR and Toxicities

- ❖ NNRTI agents tend to be associated with
 - gastrointestinal intolerance and
 - skin rash,
 - infrequently be serious (eg, Stevens-Johnson syndrome).
- ❖ **Metabolized by the CYP450 system**
 - potential drug-drug interactions
- ❖ All NNRTI agents are substrates for CYP3A4 and can act as
 - ✓ Inducers.... (nevirapine),
 - ✓ Inhibitors..... delavirdine
 - ✓ mixed inducers and inhibitors.... (efavirenz, etravirine).

- ❖ Delavirdine is a bis-heteroaryl piperazine NNRTI that selectively inhibits HIV-1

PK

- ❖ oral bioavailability85%, It is
- ❖ extensively bound (~98%) to plasma proteins and has correspondingly low cerebrospinal fluid levels.
- ❖ Serum half-life is 6 hours.

Adverse effect and toxicities

- ❖ **Skin rash (38%)**
 - occurs during the first 1–3 weeks of therapy and does not preclude rechallenge.
 - severe rash such as erythema multiforme and Stevens-Johnson syndrome.
- ❖ **Other possible adverse effects**
 - headache, fatigue, nausea, diarrhea, and increased serum aminotransferase levels.

Drug Interaction

- ❖ Delavirdine is extensively metabolized by CYP3A and CYP2D6 enzymes,
- ❖ **rifampin** or **rifabutin** are also contraindicated.
- ❖ **Delavirdine** + indinavir / saquinavir
 - prolongs the elimination half-life of (**BID** than **TID** is suffice)
- ❖ **Bioavailability**
 - reduced by antacids or H2-blockers.

The pyridinone doravirine may be taken **with** or **without food**.

PK

- ❖ bioavailability of doravirine64%
- ❖ Protein binding.....76% and half-life.....15 hours.

Adverse Effects

most common adverse effects

- ✓ headache, nausea, diarrhea, and rash.
- ❖ **Compared to Efavirine and ritonavir-boosted darunavir.**
- ✓ Doravirine has **fewer CNS side effects** and **fewer negative effects on lipid profile**

Doravirine...

Drug interaction

- ❖ Doravirine is metabolized by hepatic cytochrome P450
- ✓ **Drugs**rifampin, rifapentine, efavirenz, certain anticonvulsants (eg, carbamazepine, phenobarbital, and phenytoin), and St. John's wort substantially decrease doravirine concentrations and are C/I .
- ❖ **doravirine + rifabutin**
 - ✓ The **dosage doravirine** of must be increased if co administered

Efavirenz

- ❖ Efavirenz has been used widely because of its
 - ✓ convenience,
 - ✓ effectiveness, and
 - ✓ long-term tolerability
- ❖ can be given once daily because of its long half-life (40–55 hours).

PK

- ❖ It is moderately well absorbed following oral administration (45%).
- ❖ **should be taken on an empty stomach.**
- ❖ **Protein bindingbound to albumin (~99%),**

Metabolism

- ❖ metabolized by **CYP3A4** and **CYP2B6** to inactive hydroxylated metabolites;
- ❖ the remainder is eliminated in the feces as unchanged drug

❖ **The principal adverse involve the CNS.**

- Dizziness, drowsiness, insomnia, nightmares, and headache

 - tend to diminish with continued therapy;

- dosing at bedtime may also be helpful.

✓ **Psychiatric symptoms** such as depression, mania, and psychosis **following initiation** and may **necessitate discontinuation.**

✓ **mild-to-moderate skin rash (28 %)**

- Resolve despite D/C

- Rarely, rash has been severe or life-threatening.

Adverse effects

❖ Other potential adverse reactions

- ✓ nausea, vomiting, diarrhea, crystalluria, elevated liver enzymes, and increased total serum cholesterol by 10–20%.

❖ **Unlike in animal, fetal tube defect** is not observed in human

- ✓ recommended as an alternative during pregnancy

Etravirine

- ❖ It is a diarylpyrimidine, and was designed to be effective against strains of **HIV** that had developed resistance to first-generation **NNRTIs**

PK

- ❖ **Etravirine should be taken with meal**
- ✓ Highly protein bound
- ✓ Metabolism....primarily metabolized by the liver.
- ✓ half-life is ~41 hours.

Adverse effects

The most common adverse effectsrash (mild), nausea, and diarrhea.

- ✓ rash resolves after 1–2 weeks without D/C
- ✓ Rarely, rash has been severe or life-threatening.

- ❖ Laboratory abnormalities.... elevations in serum cholesterol,
- ❖ triglyceride,
- ❖ glucose, and
- ❖ hepatic aminotransferase levels.
 - ✓ Aminotransferase elevations are more common in patients with HBV or HCV co-infection.

Etravirine....

Etravirine is a **substrate and an inducer of CYP3A4** and **an inhibitor** of CYP2C9 and CYP2C19

- ✓ has potential for numerous drug-drug interactions
- ✓ Some of the interactions are difficult to predict.
 - For example, etravirine may (-) itraconazole and ketoconazole but (+) voriconazole concentrations.

Nevirapine

- ✓ Nevirapine is a dipyrindodiazepinone NNRTI
- ✓ It is approved for use in infants and children 15 days old or older

Uses

- ❖ In the prevention of transmission of HIV from mother to newborn

Administration.... A single dose 200 mg at the onset of labor and followed by a 2-mg/kg dose to the neonate **within 3 days** of delivery.

- ✓ no longer recommended for use in pregnancy

PK

- ✓ **Oral BA>90% and absorption is not dependent with food**
- ✓ Serum half-life....25–30 hours. It is
- ✓ extensively metabolized by the CYP3A
- ✓ It is highly lipophilic

Adverse Effects

- ❖ Rash (20%), that spares the palms and soles,
 - ✓ usually in the first 4–6 weeks of therapy.
 - ✓ **Gradual dose escalation over 14 days** to decrease the frequency of rash.
 - ✓ The incidence of rash is greater in women
 - ✓ **Severe rash** is rare but **greater than other NNRTIs**
- ❖ **Symptomatic liver toxicity may occur in up to 4%**
 - ✓ Fulminant, life-threatening hepatitis
 - within the first 18 weeks of therapy.
- ❖ Other adverse effects... fever, nausea, headache, and somnolence.

Drug interaction

Nevirapine is a moderate inducer of CYP3A metabolism,

Rilpivirine

- ❖ Rilpivirine, a diarylpyrimidine,
- ❖ **must be administered with a meal** (preferably high fat or >400 kcal).
- ❖ **Requires Acidic environment for absorption**
 - ✓ antacids and H₂-receptor antagonists should be separated in time and
 - ✓ PPIs are C/I.
- ❖ It is highly protein-bound and
- ❖ half-life.....50 hours.
- ❖ **Used during pregnancy**

Adverse Effects

❖ The most common

–rash, depression, headache, insomnia, and increased serum aminotransferases.

❖Others

✓Increased serum cholesterol and fat redistribution

✓QT prolongation,

–coadministration with agents that may prolong the QT interval should be avoided.

Drug interaction

✓metabolized by CYP3A4

–Coadministration of rifampin or rifabutin is contraindicated.

Protease Inhibitors (PIs)

- ❖ During the later stages of the HIV the *gag* and *gag-pol* genes....translated into **polyproteins**
 - It become budding particle
- ❖ **The HIV protease** is responsible for cleaving these precursor molecules
 - produce the final structural proteins of the mature virion core.
- ❖ protease inhibitors (PIs) preventing post-translational cleavage of the Gag-Pol polyprotein,
 - prevent the processing of viral proteins into functional conformations
 - resulting in the production of immature, noninfectious viral particles.
- ❖ Unlike the NRTIs, PIs do not need intracellular activation.

Protease Inhibitors (PIs)...

Adverse effects

- ❖ PIs are associated with
 - gastrointestinal intolerance, and
 - lipodystrophy,
 - both metabolic (hyperglycemia, hyperlipidemia) and
 - morphologic (lipoatrophy, fat deposition) derangements.
- ❖ A syndrome of redistribution and accumulation of body fat
 - central obesity,
 - dorsocervical fat enlargement (buffalo hump),
- ❖ peripheral and facial wasting
- ❖ breast enlargement
- ❖ cushingoid appearance

Adverse effects

- ❖ PIs may be associated with cardiac conduction abnormalities, including **PR** and **QT** interval prolongation.
 - A baseline electrocardiogram (required) and
 - avoidance of other agents causing prolonged PR or QT intervals
- ❖ Drug-induced hepatitis and rare severe hepatotoxicity
 - more in **tipranavir/ritonavir** than with other PIs
- ❖ Unconjugated hyperbilirubinemia may occur with **atazanavir** or **indinavir**.
- ❖ **increased spontaneous** bleeding in patients with **hemophilia**

Protease Inhibitors (PIs)...

- ❖ Increased risk of intracranial hemorrhage..in tipranavir/ritonavir.
- ❖ Darunavir, amprenavir, fosamprenavir, and tipranavir are *sulfonamides*;
 - caution should be used in patients with a history of sulfa allergy.

Drug interaction

- ❖ **All of the antiretroviral PIs** are extensively metabolized by CYP3A4
- ❖ **Ritonavir (most) and saquinavir has (the least)** inhibitory effect
 - ✓ ritonavir are used to clinical advantage by having it “boost” the levels of other PI agents...used in combination
 - ✓ Serve as pharmacokinetic enhancer rather than an antiretroviral agent
- ❖ **amprenavir and ritonavir**, are also inducers of specific CYP isoforms

- ❖ Atazanavir is an azapeptide PI
- ❖ Boosted atazanavir can be given during pregnancy

PK

- ❖ requires an acidic medium for absorption and
 - Separation of ingestion from acid-reducing agents
 - PPIs are C//I..**
- ❖ half-life is **6–7** hours and **8–9** with ritonavir
- ❖ Elimination....primarily biliary
 - should not be given to patients with severe hepatic insufficiency

Adverse effects

- ❖ The most common adverse
 - diarrhea and nausea; vomiting, abdominal pain, headache, and peripheral neuropathy may also occur

Adverse Effects

- ❖ Mild Skin rash (~20%)
- ❖ severe rash and Stevens-Johnson syndrome have been reported.
- ❖ indirect hyperbilirubinemia with **overt jaundice** (~10%)
 - This is also occurred in **indinavir**
 - associated with inhibition of the **UGT1A1 glucuronidation enzyme**.
- ❖ Elevation of serum aminotransferases
 - Especially in underlying HBV or HCV co-infection.
- ❖ Kidney stones, gallstones, PR prolongation, and decreased bone mineral density have also been reported..

Adverse Effects

- ❖ phenylketonuria
 - It contains phenylalanine

Atazanavir

Atazanavir is an inhibitor of CYP3A4, CYP2C9, and UGT1A1, the potential for drug-drug interactions with atazanavir is great

- ❖ unboosted atazanavir should not be administered with
 - cisapride, ergotamine, etravirine, fosamprenavir, nevirapine, PPI , or tipranavir.
- ❖drugs that inhibit **UGT1A1**, such as irinotecan, **may increase atazanavir** levels.
- ❖levels of **Atovaquone** and **voriconazole** can be **decreased** and
- ❖levels of **maraviroc** and **ranolazine** may be increased.

- ❖ Darunavir is a nonpeptidic PI that is active against both HIV-1 and HIV-2.
- ❖ binds tightly but **reversibly** to the active site of **HIV protease**
 - ✓ **Can also** prevent protease dimerization
- ❖ Boosted darunavir is one of the PI agents recommended for use in **pregnancy**

PK and drug interaction

- ❖ Should be **taken with meals** to improve **bioavailability**
- ❖ must be given with **ritonavir** or **cobicistat**.
- ❖ It is highly **protein-bound** and primarily **metabolized by the liver**.

Drug interaction

- ❖ both inhibits and is metabolized by the CYP3A enzyme system,
 - Drug interaction by is compounded by Ritonavir (booster)
 - potent inhibitor of CYP3A and CYP2D6,
 - inducer of other hepatic enzyme systems.
- ❖ Levels of clarithromycin, colchicine, cyclophosphamide, and digoxin ...(+)
- ❖ and levels of **paroxetine** and **sertraline** may be **decreased**.

Adverse Effects

- ❖ diarrhea, nausea, headache, and
 - increases in amylase and hepatic aminotransferase levels.
 - Liver toxicity, including severe hepatitis
 - liver function tests should be monitored
 - be higher for persons with HBV, HCV, or other chronic liver disease
- ❖ Rash (2–7%)
- ❖ contains a sulfonamide moiety
 - can cause a hypersensitivity reaction...with sulfa allergy

❖ Fosamprenavir is a **prodrug** of **amprenavir**

–hydrolyzed by enzymes in the intestinal epithelium

❖ administered in combination with low-dose ritonavir.

❖ It can be taken with or without food.

–But **high-fat meals decrease absorption** and **should be avoided.**

❖ half-life is relatively long (7–11 hours).

❖ Amprenavir is metabolized in the liver and

–should be used with caution in the setting of hepatic insufficiency.

Adverse Effects

❖ The **most common** adverse effects

–headache, nausea, diarrhea, perioral paresthesias, and depression.

❖ The oral suspension, which **contains propylene glycol**,

–C/I in young children pregnant women, patients with renal or hepatic failure, and those using **metronidazole** or **disulfiram**.

❖ contains a sulfa moietyRash (19%)

–**Sometimes need to D/C**

Drug interaction

❖ Amprenavir is both an inducer and an inhibitor of CYP3A4

–Coadministration of atazanavir, etravirine, lopinavir/ritonavir, posaconazole, and ranolazine is C/I

Drug interaction...

❖ **amprenavir + ritonavir.....oral solutions** should not be coadministered

—propylene glycol and the ethanol compete for the same metabolic pathway,

• **Accumulation of either.**

❖ oral solution contains **vitamin E** at several times the recommended daily dosage,

—supplemental vitamin E should be avoided.

- ❖ Indinavir, a peptidomimetic HIV PI,
- ❖ lacks significant advantages over other HIV PIs
 - no longer widely prescribed
 - because of problems **nephrolithiasis** and other **nephrotoxicities**

PK

- ❖requires an acidic environment for optimum
- ❖half-life is1.5–2 hours,
- ❖protein binding.....~60%,
- ❖Excretion is primarily fecal.

Adverse Effects

The most common

– **unconjugated hyperbilirubinemia and nephrolithiasis**
10%.)

- associated with urinary crystallization of the drug.
- Nephrolithiasisstarts days after initiating therapy,
- Acute renal failure and interstitial fibrosis

❖ at least 48 ounces of water daily is important
– to maintain adequate hydration

❖ Nausea, diarrhea, sicca syndrome, headache, blurred vision, and

❖ elevations of serum aminotransferase

❖ Insulin resistance

– **common with indinavir** than with the other PIs,
occurring in 3–5% of patients.

Drug interaction

- ❖ Indinavir is inhibitor of CYP3A4, numerous and complex drug interactions can occur
- ❖ Boosting with ritonavir
 - ✓ allows for twice-daily than TID
 - ✓ But also potential for an increase in **nephrolithiasis**
 - thus, a high fluid intake (1.5–2 L/d) is advised.
- ❖ Indinavir **should not be coadministered** with
 - ✓ astemizole, cerivastatin, efavirenz, ergotamine, etravirine, lovastatin, pimozide, rifampin, simvastatin, terfenadine, or triazolam.
 - ✓ **Levels of** amlodipine, levodopa, and trazodone....(+)

- ❖ It recommended antiretroviral agents for use in pregnant women
- Used as “booster

PK

- ✓ bioavailability (~75%) that **increases with food.**
- ✓ Protein binding.....98% protein-bound
- ✓ half-life.....3–5 hours.
- ✓ Metabolism to an active metabolite occurs by (CYP3A and CYP2D6)
- ✓ excretion is primarily in the feces.

Full-dose ritonavir include

- ✓ asthenia, gastrointestinal disturbances, and hepatitis;
 - Reduce **by lowering the booster dose** and by
 - Dose escalation over **1–2 weeks** decreases these adverse effects

Other potential adverse effects include

- ✓ altered taste, headache
- ✓ paresthesia (circumoral or peripheral),
- ✓ elevated serum aminotransferase and lipid levels,
- ✓ elevations in serum creatine kinase, and pancreatitis

Drug interaction

- ❖ It is a potent inhibitor of CYP3A4
 - has been used to great advantage when ritonavir is administered in low doses (100–200 mg twice daily)
- ❖ in combination with any of the other PI agents,
 - permit lower or less frequent dosing (or both)
 - Tend to be tolerable
 - Potential for greater efficacy against resistant virus.
- ❖ **digoxin** and **theophylline** should be monitored
- ❖ **saquinavir + ritonavir**.....C/I (for whom at risk)
 - increased risk of **QT prolongation** and **PR interval prolongation**.

Saquinavir

- ❖ Saquinavir is a peptidomimetic hydroxyethylamine

PK and the effect food

- ❖ Should be taken within **2 hours after a fatty meal**
- ❖ protein-bound.....97%,
- ❖ half-life2 hours
- ❖ High VD but CSF penetration is negligible
- ❖ Excretionprimarily in the feces.

Adverse Effects

- ❖ Gastrointestinal discomfort (nausea, diarrhea, abdominal discomfort, dyspepsia) may occur.
- ❖ **less dyslipidemia or gastrointestinal toxicity**
 - low-dose ritonavir + Saquinavir compared other boosted PIs
- ❖ Prolongation of the **QT interval** and **torsades de pointes**

Adverse Effects

- should not be used in patients with congenital long QT syndrome, AV block, and refractory hypokalemia
- in combination with drugs that prolong the QT interval.

Drug interaction

- ❖ metabolism by CYP3A4 and inhibits CYP3A4 inhibitor as
- ❖ Digoxin levels should be monitored.
- ❖ **Liver tests** should be monitored if saquinavir is coadministered with **delavirdine** or **rifampin**.
- ❖ darunavir or tipranavir..... **C/I**

- ❖ Tipranavir is a newer PI used in patients who has resistant to other PIs
 - used in combination with ritonavir
- ❖ Bioavailability is poorcan be increased a **high-fat meal**
- ❖ It is C/I in patients with hepatic insufficiency.
- ❖ should not be administered to patients with known **sulfa allergy**

Adverse Effects

Most common adverse effects

- ❖ diarrhea, nausea, vomiting, and abdominal pain
- ✓ urticarial or maculopapular rash (10–14%).....common with **ethinyl estradiol**

Liver toxicity, including hepatic decompensation,

—more common than with other **PIs**

❖ an increased **risk for intracranial hemorrhage** in patients with head trauma or bleeding diathesis

—**tipranavir/ritonavir**....should be avoided

❖ **Other potential adverse effects**

✓ depression,

✓ elevation in serum amylase,

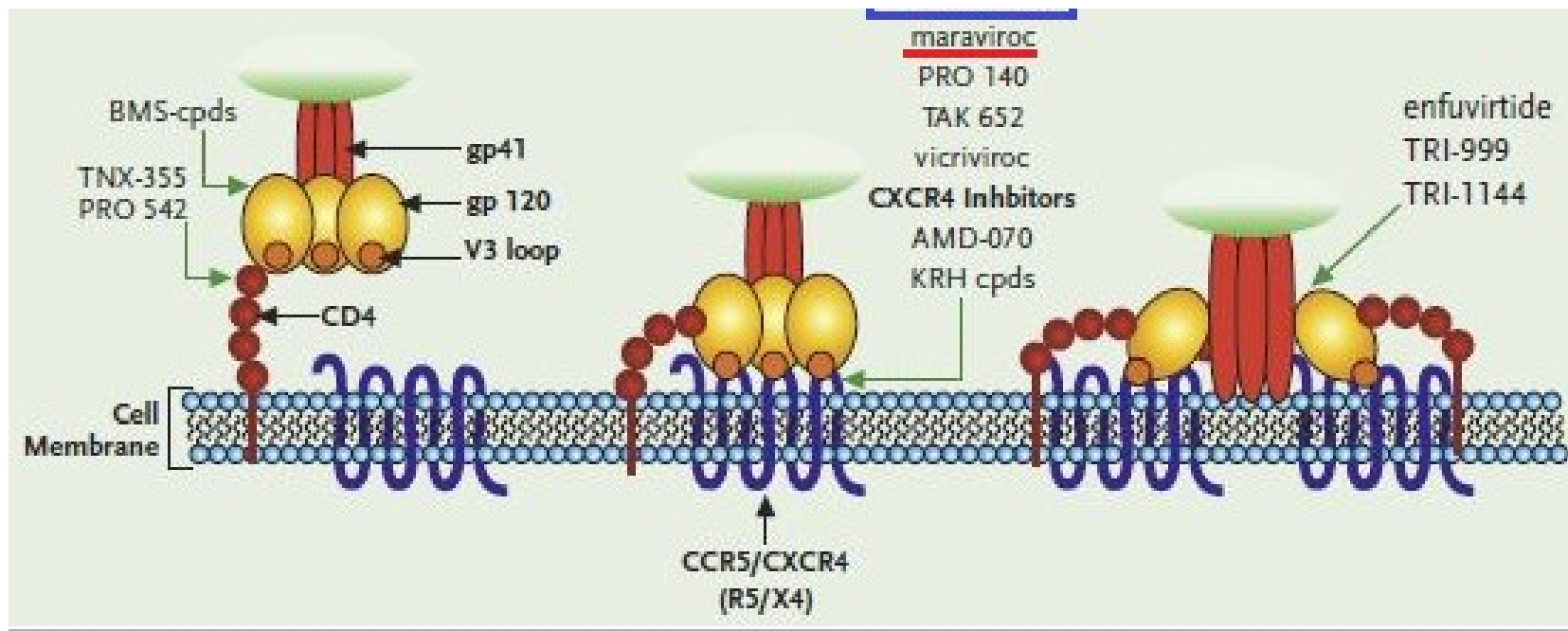
✓ increased serum lipids, and

✓ decreased white blood cell count.

Drug interaction

- ❖ Tipranavir both inhibits and induces the CYP3A4 system.
 - The **net effect is inhibition** when combined with ritonavir
- ❖ Tipranavir also **induces the P-glycoprotein transporter**
- ❖ Tipranavir with one of the following should be avoided
 - atazanavir, elvitegravir/cobicistat, etravirine, fosamprenavir, lopinavir/ritonavir, and saquinavir
- ❖ **vitamin E** is contraindicated in patients receiving the oral solution.

- ❖ The process of HIV-1 entry into host cells is complex;
 - each step presents a potential target for inhibition.
- ❖ Viral attachment to the host cell
 - binding of the viral envelope glycoprotein complex gp160 (consisting of gp120 and gp41) to its cellular receptor CD4.
 - binding induces conformational changes in gp120 that enable access to the chemokine receptors **CCR5** or **CXCR4**.
 - Chemokine receptor binding induces further conformational changes in gp120,
 - allowing exposure to gp41 and leading to fusion of the viral envelope with the host cell membrane and subsequent entry of the viral core into the cellular cytoplasm.



- "A schematic illustration of HIV entry in the steps presented by

Figure: Entry and Fusion

- ❖ It is a synthetic 36-amino-acid peptide fusion inhibitor that blocks HIV entry into the cell

MOA....Binds to the **gp41** subunit of the **viral envelope** glycoprotein,

–Prevent the conformational changes required for the **fusion of the viral** and cellular membranes.

✓ **Administered via SC...**

PK

❖ Half-life..... 3.8 hours.

❖ Metabolism appears to be..... proteolytic hydrolysis

Enfuvirtide....

Mechanisms of drug resistance

- Resistance results from mutations in gp41
- **No cross-resistance and interaction** with the other currently approved antiretroviral drug classes.

Adverse Effects

The most commonlocal injection site reactions.....Mild to moderate and rarely lead to discontinuation

❖ Other potential adverse effects include

–insomnia, headache, dizziness, and nausea.

❖Hypersensitivity reactionsrarely occur, are of varying severity, and may **recur on rechallenge.**

Ibalizumab

- ❖ Ibalizumab (TNX-355) is a monoclonal antibody that
- ❖ **MoA...** binds the **CD4 receptor** and **blocks** HIV entry to CD4 cells.
- ❖ It is indicated multidrug-resistant HIV
- ❖ It is administered IV.
- ❖ must be combined with additional active agents

Most common ADE

- ✓ diarrhea, dizziness, nausea, and rash.

Maraviroc

MOA.....Maraviroc blocks the binding of the HIV outer envelope protein gp120 to the CCR5 chemokine receptor

Maraviroc is approved

- ✓ adults who have baseline evidence of predominantly CCR5-tropic virus.

Mechanisms of resistance

- ✓ **A shift in tropism (from CCR5 to CXCR4)**
- ✓ - or dual/mixed tropism predominance.
- ✓ mutations of gp120

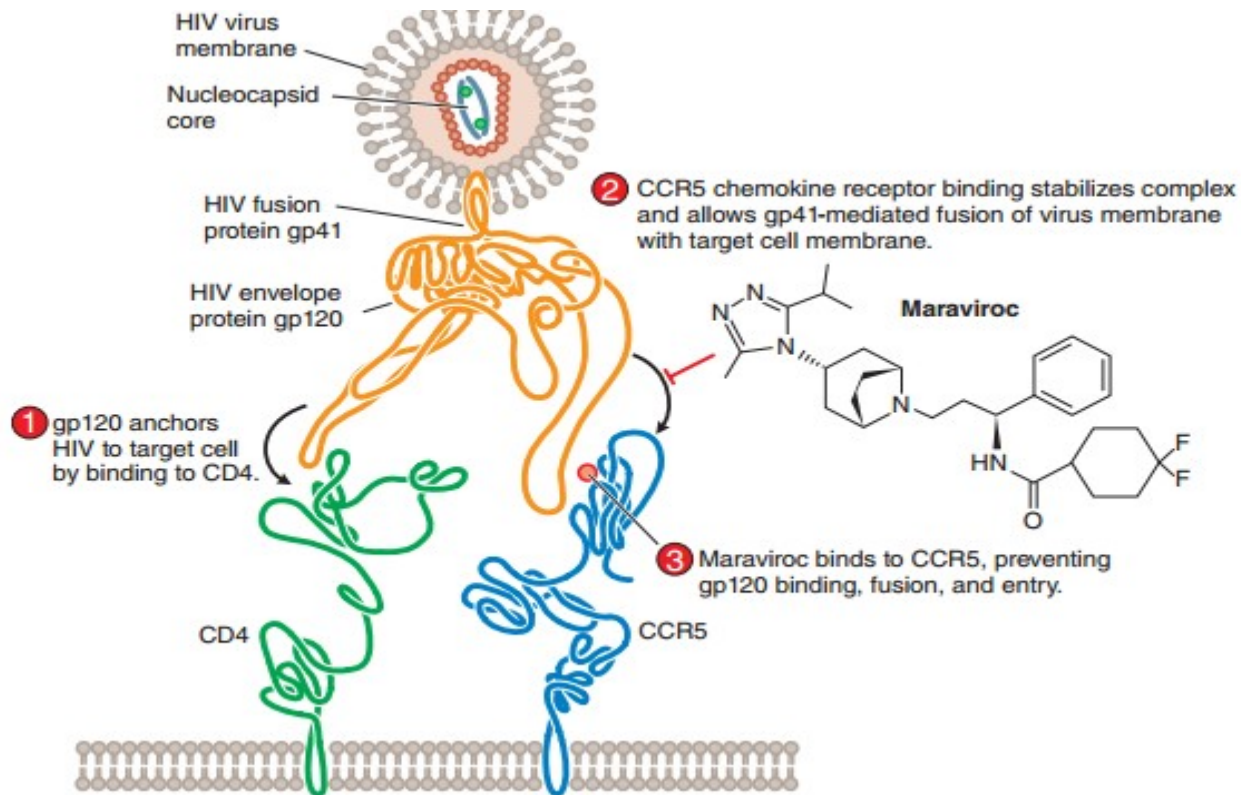


Figure: Mechanism of action of the HIV entry inhibitor maraviroc

Maraviroc

It is the only antiretroviral drug approved at three doses

- ✓ 150 mg BID with CYP3A inhibitors
- ✓ with most CYP3A inducers.....600 mg BID
- ✓ Drugs with no significant inhibition/induction....300 mg BID
- ✓ **Oral BA...23%–33% (food decreases)**

Adverse Effects and Drug interaction

- ✓ dose- and concentration-dependent orthostatic hypotension
- ✓ It is CYP3A4 substrate and susceptible to pharmacokinetic interaction

Integrase Strand Transfer Inhibitors (INSTIs)

Drugs in this class: Raltegravir, Elvitegravir, Dolutegravir

MOA.... The HIV integrase inhibitor prevent formation of covalent bonds between host and viral DNA (strand-transfer)

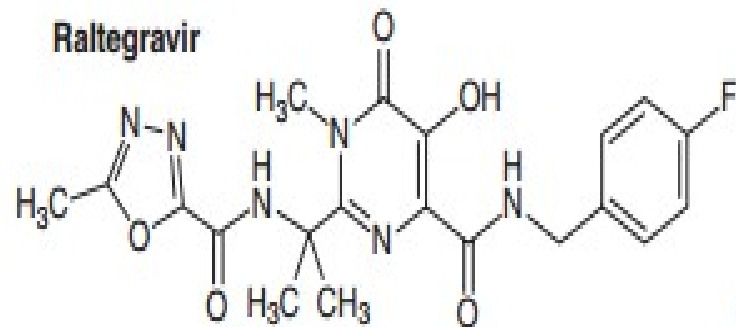
- Integrase is essential for replication of both HIV-1 and HIV-2

- ❖ As a class, these agents tend to be well tolerated but have

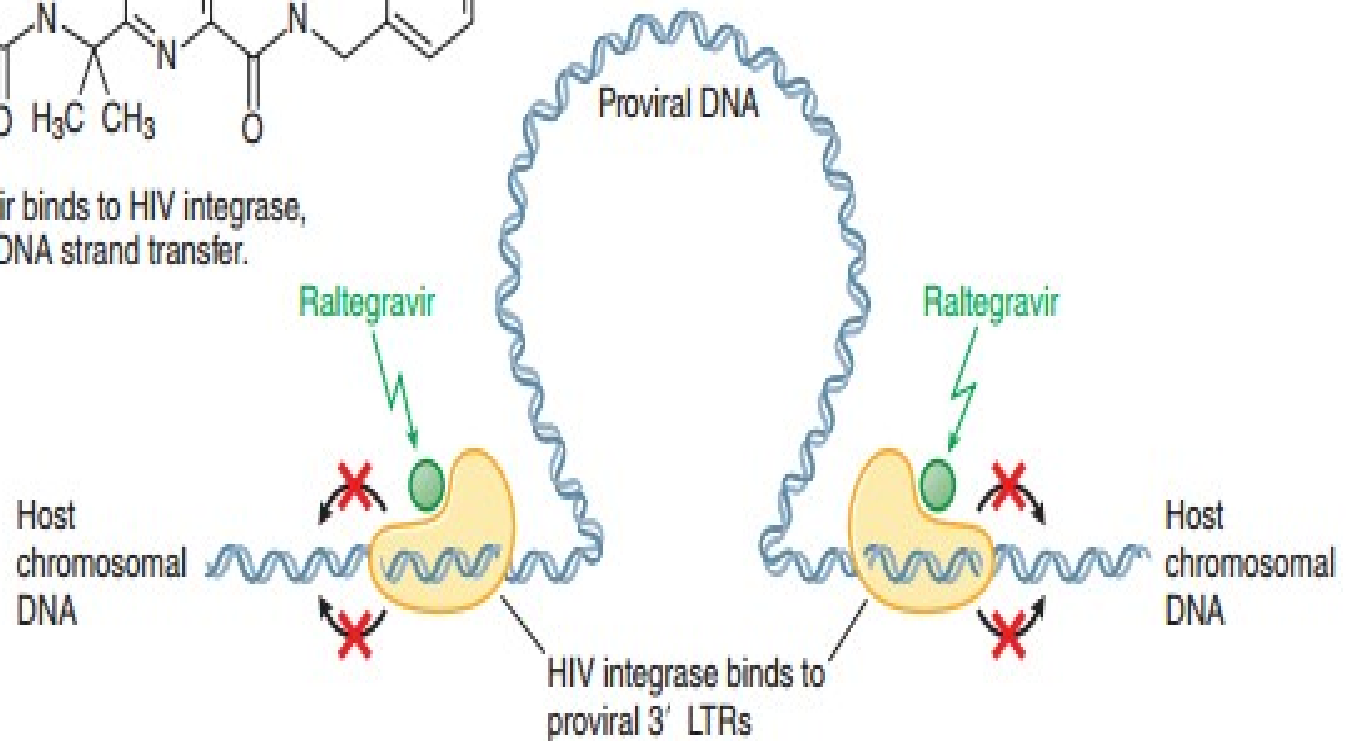
- Most common ADR..... **headache** and **gastrointestinal S/Es**

- ❖ INSTIs (eg, dolutegravir and bictegravir) have a high genetic barrier to resistance

- frequent choices as part of first-line regimens.



Raltegravir binds to HIV integrase, prevents DNA strand transfer.



Bictegravir

- ❖ available only as a **co-formulated** tablet with and given QD
- ❖ tenofovir alafenamide and emtricitabine, administered once daily.

Drug interaction and ADEs

- ❖ The most common adverse effects are
 - nausea, diarrhea, and headache.
- ❖ **cations containing drug/supplement** interfere with GI absorption of
 - should thus be **separated by at least 2 hours from ingestion of bictegravir.**

Bictegravir

It is a substrate of both CYP3A4 and UGT 1A1

–Rifampin, rifapentine, carbamazepine, phenytoin, phenobarbital, and St. John's wort ma.....decrease bictegravir plasma levels and

–**Atazanavir** may greatly increase **bictegravir** plasma levels.

❖ **Bictegravir also inhibits**

– the renal transporters organic cation transporter 2 (OCT2) and multidrug and **toxin extrusion transporter 1** (MATE1)

•may increase levels of drugs that are substrates of these transporters, such as metformin.

Dolutegravir

- ❖ Dolutegravir interacts with antacids or laxatives, sucralfate, oral iron supplements, oral calcium supplements
 - should be taken 2 hours before or 6 hours after cation-containing, or buffered medications.

PK

- Peak plasma.....2–3 hours of ingestion.
- Protein binding.... 99%
- half-life is ~14 hours.

Dolutegravir....

- Dolutegravir retains activity against
 - viruses that have become resistant to antiretroviral agents of other classes and also
 - against some viruses that are resistant to raltegravir or elvitegravir.

Dolutegravir

- ❖ Dolutegravir is approved for
 - HIV-infected adults and children more than 12 years of age and weighing at least 40 kg.
- ❖ Available as a co-formulation
 - ABC+ 3TC+lamivudine+Dolutegravir
- ❖ It is highly efficacious, tolerability, and convenience

- ❖ ADR are infrequent and includes include
 - insomnia, headache, increased serum aminotransferase levels, and rarely, rash
- ❖ **A hypersensitivity** reaction, including rash and systemic symptoms
 - **Should be D/C and do not be re-initiated**
- ❖ Increases serum creatinine
- ❖ **neural tube defects** among infants born to women taking dolutegravir-containing regimens
 - Should be avoided in first but drug of choice after **1st trimester compared to other drugs within this class**

Drug ineteraction

- **Metabolized priamarly** via UGT1A1 with some contribution from CYP3A.
- ❖ It inhibits the renal organic cation transporter **OCT2**,
 - Increase drug concentration which are substrate for OCT2
 - **dofetilide** is C/I and close monitoring if Metformin co-administered

Elvitegravir

- ❖retains activity against viruses that have become resistant to antiretroviral agents of other classes,
 - but shares** cross-resistance with mutations affecting the **HIV integrase inhibitor**

Elvitegravir....

- Available only in combination with
 - cobicistat, emtricitabine, and tenofovir disoproxil fumarate or tenofovir alafenamide
 - the combination approved for adults and children more than 12 years of age.

Drug interaction

- ❖ should be **taken with food**, and it
- ❖ should be taken **2 hours** before or **6 hours** after cation-containing drugs/supplements

- ❖ Raltegravir is pyrimidinone analog
- ❖ Is not food-dependent.
- ❖ does not interact with the cytochrome P450 system
- ❖ metabolized by glucuronidation, particularly UGT1A1.
- ❖ The chewable tablets contain **phenylalanine**
 - harmful to patients with phenylketonuria.
- ❖ has a **lower barrier to resistance** not frequently recommended
 - dolutegravir and bictegravir are preferred

Adverse effects

- ❖ Adverse effects of raltegravir are **uncommon** but include
 - nausea, headache, fatigue, muscle aches, and increased serum amylase and aminotransferase levels.
- ❖ **Severe, potentially life-threatening and fatal skin reactions.**

Experimental agents

- ❖ **Fostemsavir** binds the viral envelope protein **gp120** on **HIV-1**, adjacent to the gp120-CD4+ binding site
- ❖ it prevents the conformational change required for attachment of HIV-1 to the host cell.
- ❖ active regardless of viral tropism, without cross-resistance to any of the existing ARV compounds.
- ❖ **Cabotegravir** is a long-acting, injectable integrase inhibitor
 - Currently under investigation in combination with rilpivirine
 - the combination is administered **IM once per month**.

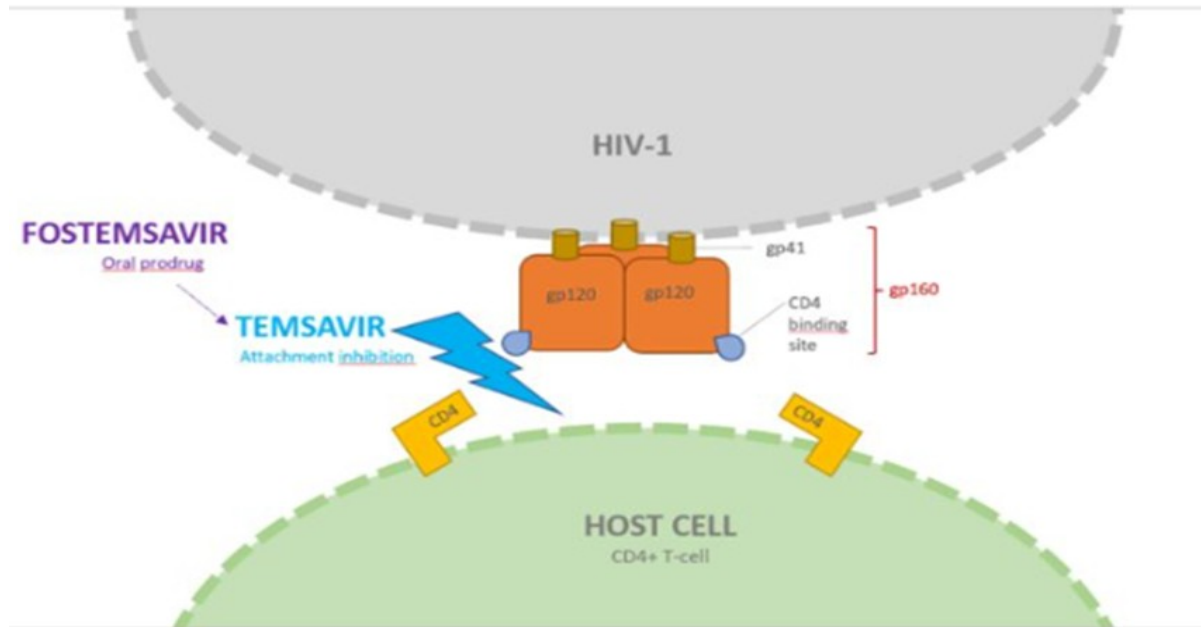


Figure: MOA of Fostemsavir